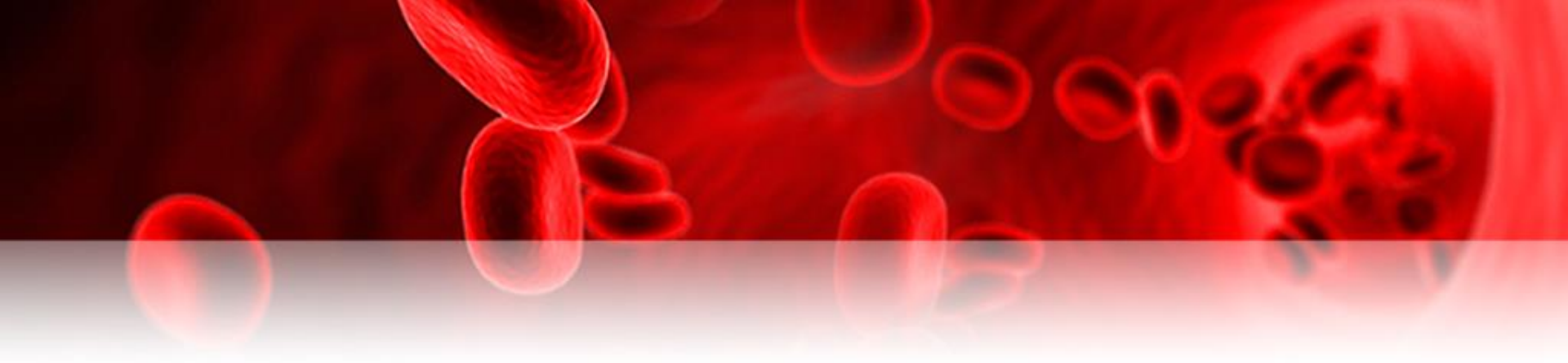


Estimation of hemoglobin content

Sahli Apparatus/Haemometer

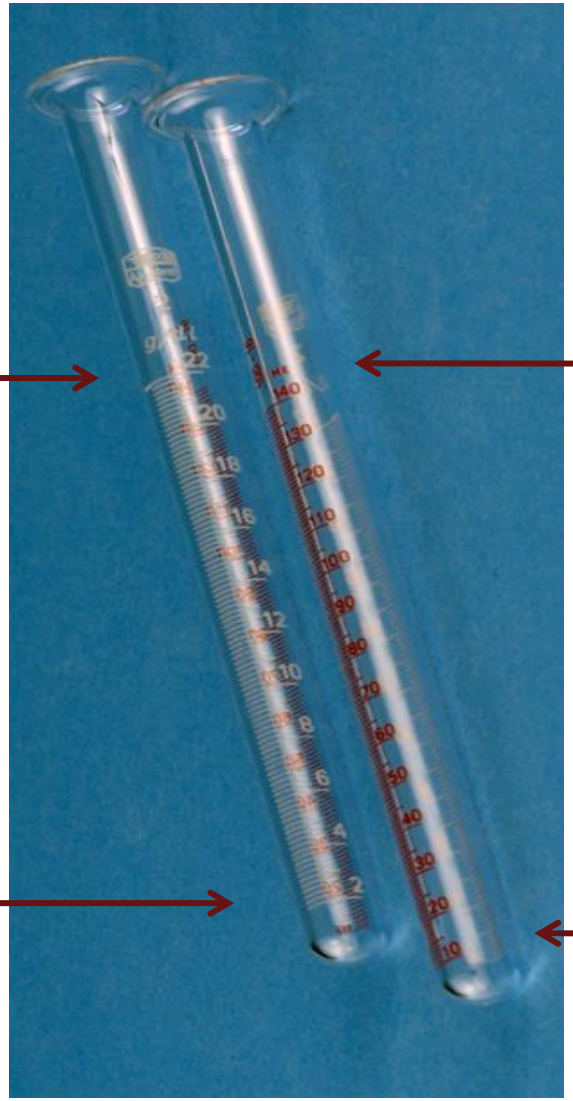

MARIENFELD
Haemometer
Haemometer acc. to Sahli, complete set consisting of:
polystyrene support with 2 colored rods and opal glass
plate, comparator tube, haemoglobin pipette 20 µl with
dropping pipette
clearing brush
Made in Germany
32 430 00





Lateral standard tubes containing standard amount of acid hematin from blood of 15 g % Hb and their color is stable.

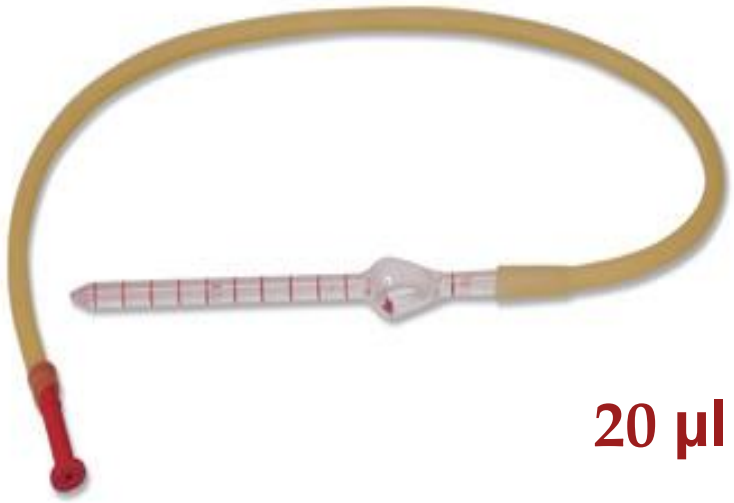
Haemometer Tube



Blood pipette







20 µl = 20 mm^3 = 0.02 ml

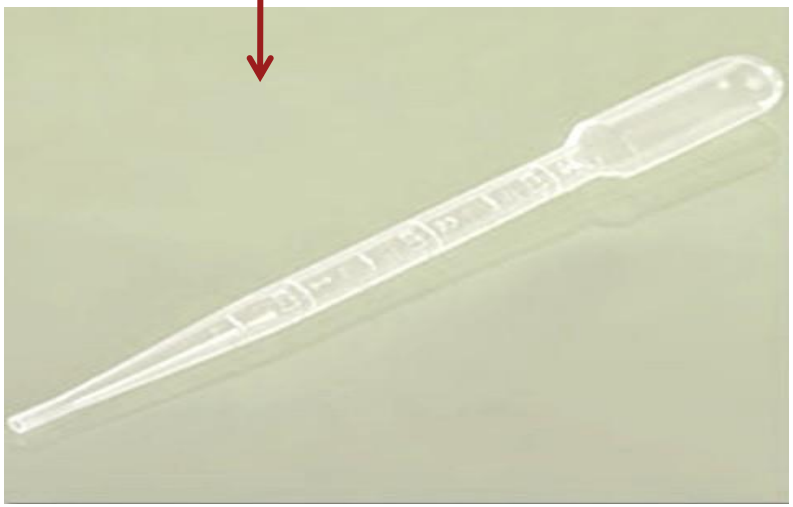
Distilled water

Blood



HCL

Dropper



Principle

- Conversion of hemoglobin to acid hematin and comparing visually the colour developed with that standard glass (colouremetric method).

Procedure

1-Fill the haemometer tube with diluted HCl till the level **10%** graduation of the tube.

2- Suck blood into the capillary tube until 20 cmm mark is reached.

3- Blow blood out of the capillary pipette into HCl haemometer.

4- Mix the blood with the acid solution by a glass rod by rotating and gently moving up and down.

5- Put the haemometer tube in the comparator and leave for 5 minutes.

6- Take the Hb tube out of the comparator and distilled water drop by drop stirring the mixture each time and comparing the color with the standard.

7- Hold the comparator at eye level away from your face against bright but diffused light.

8- Read against both scales, the Hb level using the g% scale and the % as normal using the % scale.

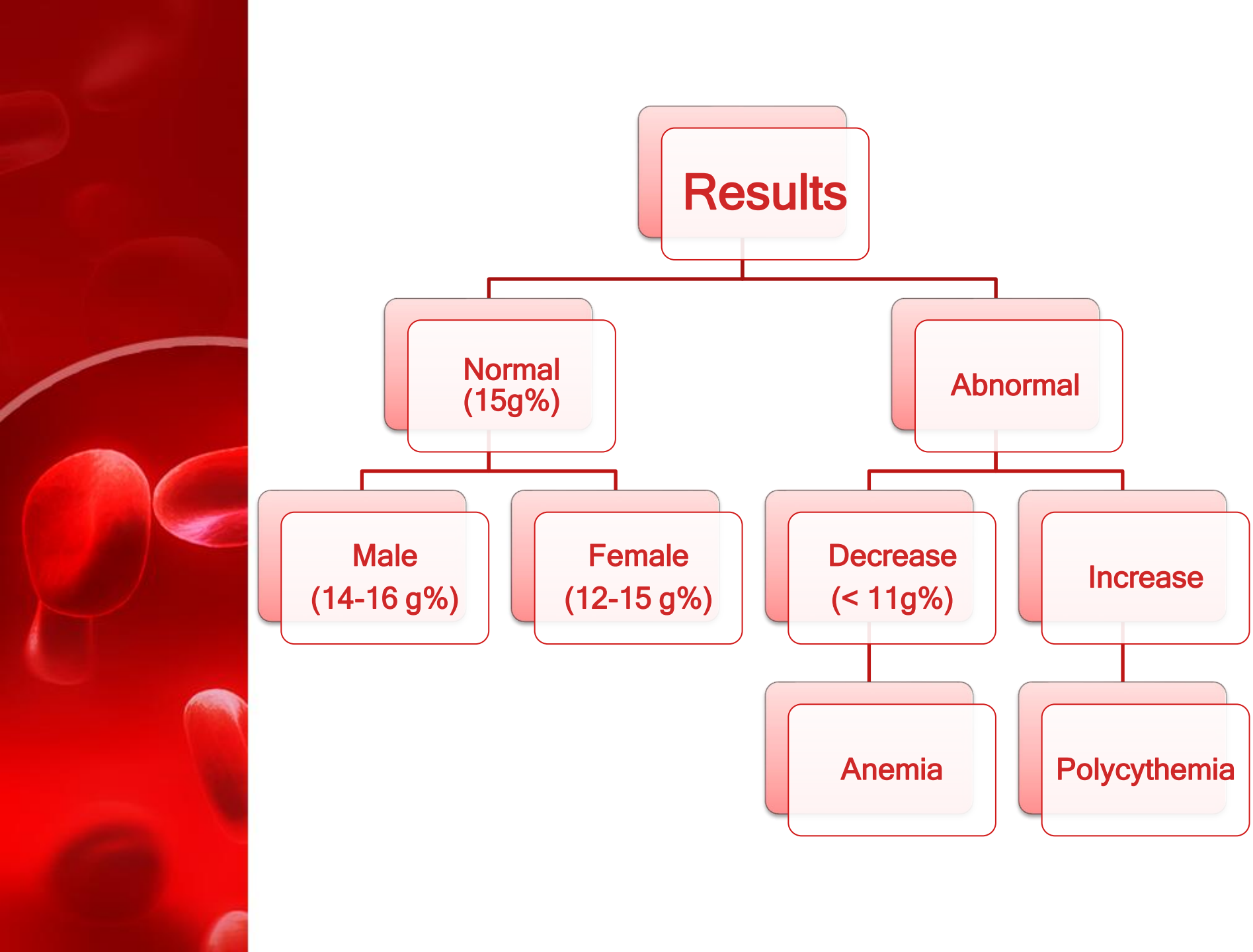

Assistent


ERKA





Read at the lower border of the fluid concave surface


Results
Normal (15g%)
Abnormal
Male (14-16 g%)
Female (12-15 g%)
Decrease (< 11g%)
Increase
Anemia
Polycythemia

• Advantage:

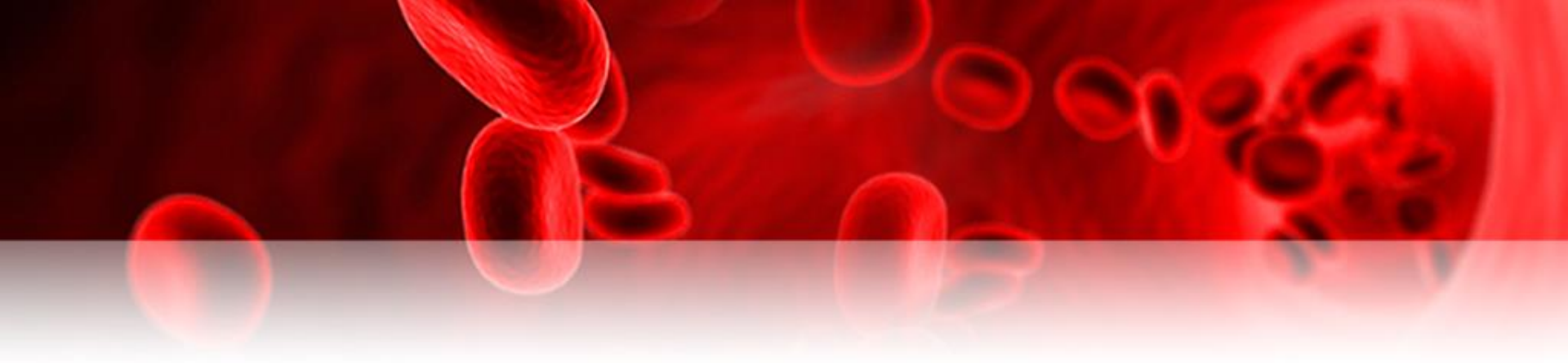
- This method is simple, does not require any costly apparatus, and can be used in mass surveys.

• Disadvantages:

- Subjective visual colour comparison.
- Need accurate pipetting.
- Estimation of only acid hematin formed.

Hemoglobin Estimation by Sahli's apparatus checklist

	Points of Evaluation	*Weight (in marks)*	Marks of the student
	Steps 1-4		
1	The student filled the haemometer tube with diluted HCl till the level of 10% graduation of the tube.	0.5	
2	The student took 20 µl of blood using the automated micropipette.	0.5	
3	The student added the blood sample to the haemometer tube.	0.5	
4	The student mixed the blood sample thoroughly with the acid solution using a glass rod.	0.5	
Total marks		**2 marks**	
	Steps 5-9		
5	The student put the haemometer tube back into the apparatus and left it for 5 minutes.		
6	The student took the haemometer tube out of the apparatus, added a drop of distilled water and stirred the mixture.	0.5	
7	The student put the haemometer tube back into the apparatus, then, compared the color in the haemometer tube with the comparator tubes by holding the apparatus at the eye level against bright light.	0.5	
8	The student repeated the previous steps (6,7), till the color in the haemometer tube matched the color in the comparator tubes.	0.5	
9	The student read against both scales, the Hb level using the g% scale and the % of normal using the % scale.	0.5	
Total marks		**Any 2 steps =2 marks**	



Hematocrit Value

Hematocrit value (Packed Cell volume)

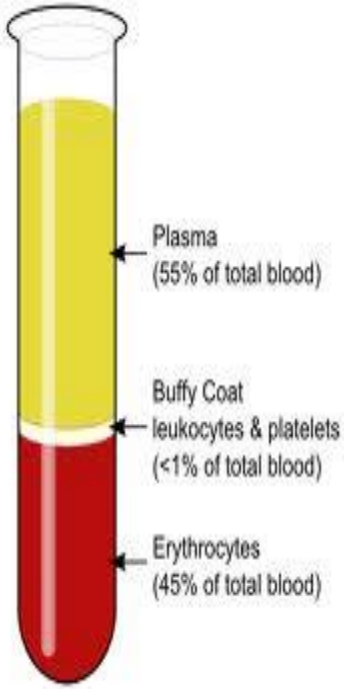
Definition:

Blood is composed of plasma and cells (RBCs , WBCs & Platelets)

since RBCs are the most abundant cells so;

Haematocrit value is defined as the volume occupied by RBCs relative to whole blood

so, it's the volume percentage of RBCs in blood



Normal values:

Male : 45 %

Female: 42 %

Children: 40 %

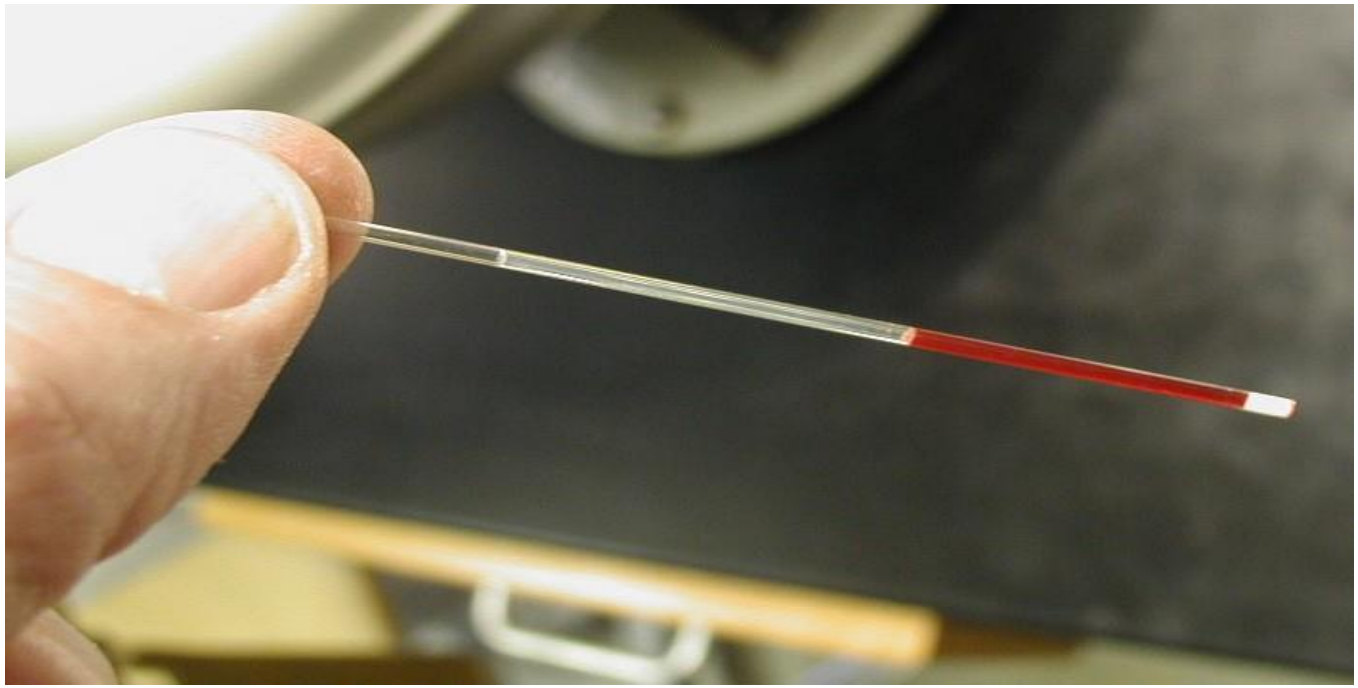
$$HV = \frac{\text{Volume of RBCs}}{\text{volume of (RBCs+WBCs+platelets+Plasma)}} \times 100$$

So

$$HV = \frac{\text{volume of RBCs}}{\text{total volume of blood}} \times 100$$

Measurement

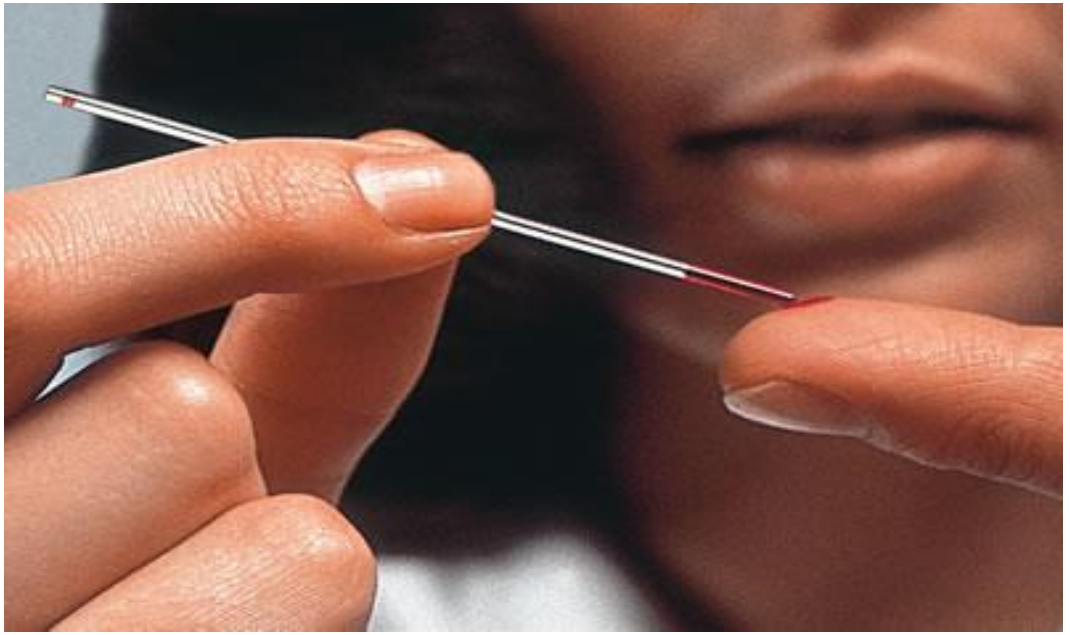
HV is measured by centrifugation of heparinized blood in a capillary tube (micro haematocrit tube)



Steps :

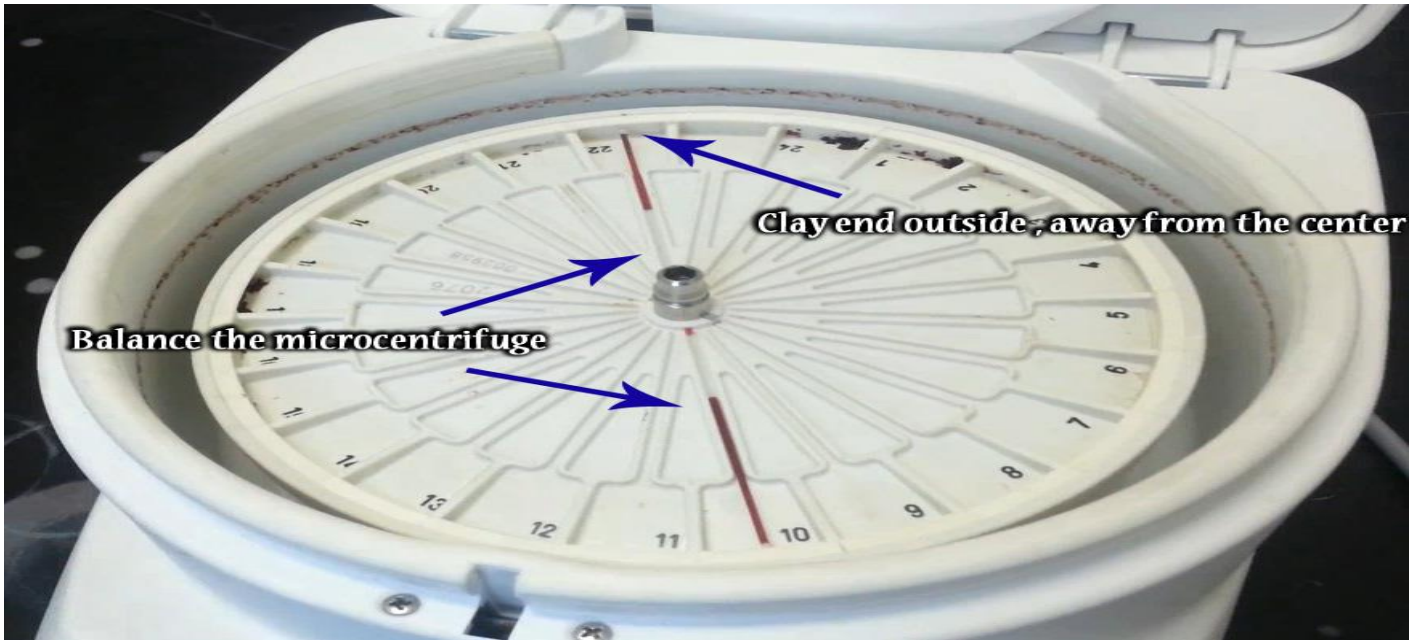
1. Clean the end of your middle finger with alcohol swab and let the finger to dry in air .
2. Remove a sterile disposable blood lancet from its package without touching the sharp end .
3. Puncture the skin on the side near the tip of middle finger with the lancet .
4. Touch a drop of blood with the end of **heparinized microhaematocrit capillary tube**

5. Hold the tube tilted slightly upwards so blood will move into the tube by capillarity .



6. Seal one end of the tube by clay at 90 °.

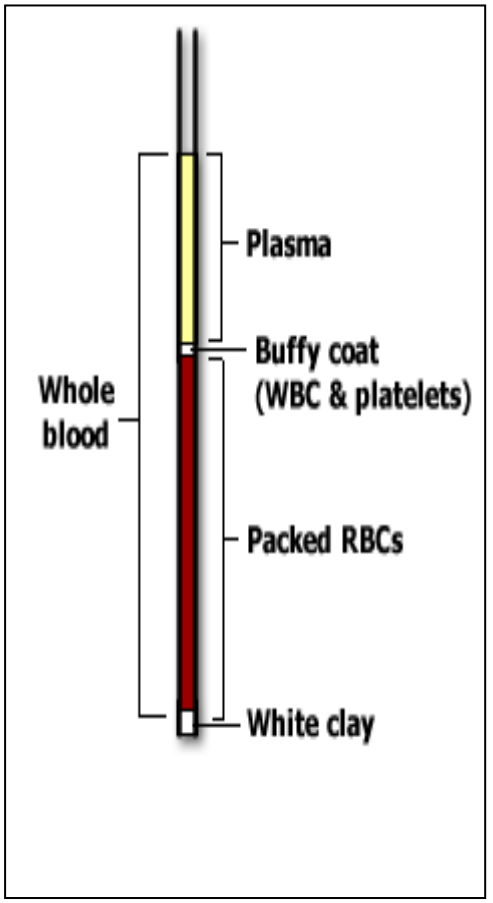
7. Place the filled and sealed capillary tubes into the centrifuge. The sealed ends should point toward the outside of the centrifuge. The duplicate samples should be placed opposite each other in order to balance the centrifuge.



8. After the centrifuge has stopped, remove the specimen tube.

The red blood cells have been packed into the bottom of the tube.

The clear liquid on top of the cells is the plasma.



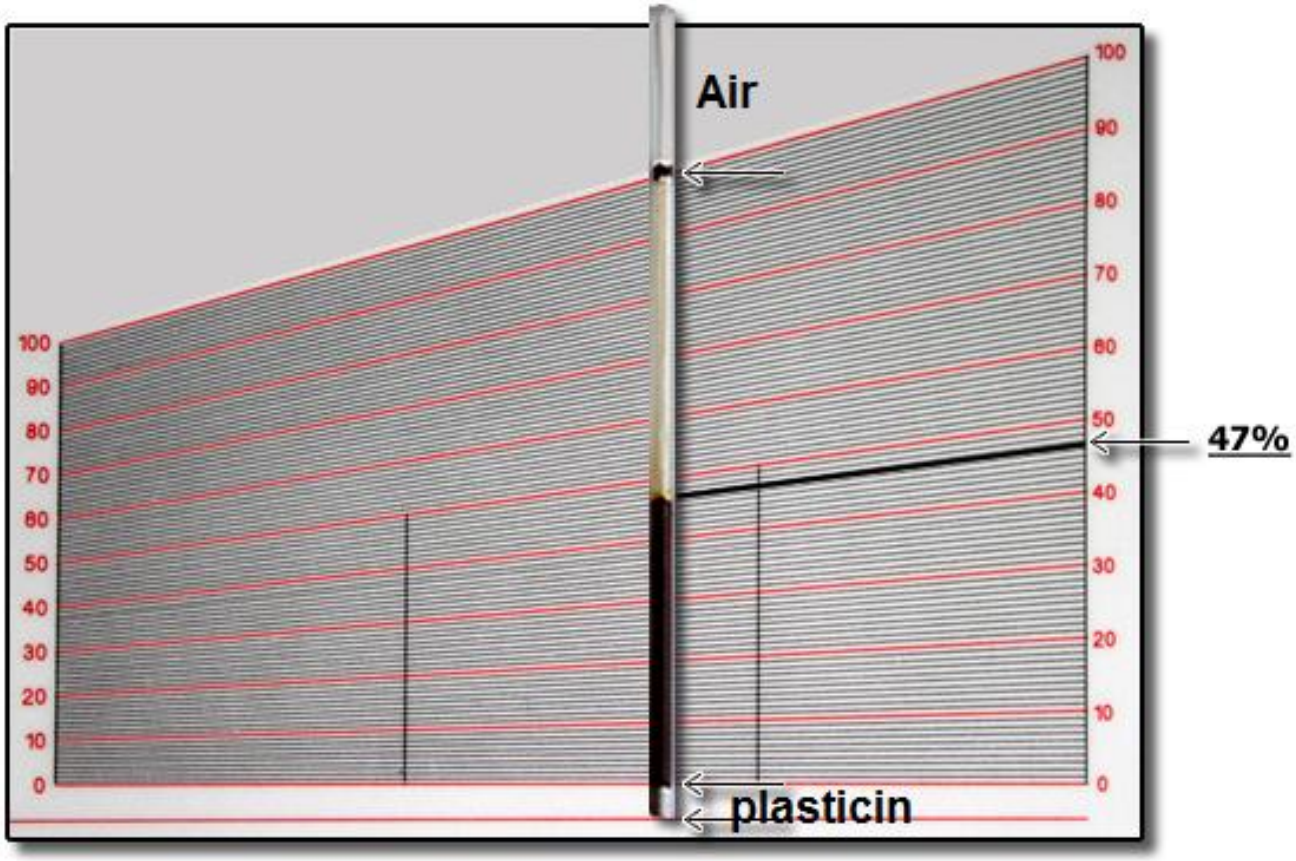
How to measure

Place the haematocrit tube on the chart such that: the lower end of the tube along the zero line of the scale such that the clay (plasticin) is below the zero line.

Move the tube along those lines until the top of the plasma intersects the top line of the scale. **N.B** :If the tube is very full, you may need to use the scale on the right side.

Then read the line where the RBCs meet the plasma

Micro haematocrit capillary tube reader



Interpretation of haematocrit value

$$\uparrow HV = \frac{\uparrow \text{volume of RBCs}}{\downarrow \text{volume of whole blood}}$$

$$\downarrow HV = \frac{\downarrow \text{volume of RBCs}}{\uparrow \text{volume of whole blood}}$$

Hematocrit value is increased in :

1.Polycythemia

2. Dehydration :

as in gastroenteritis, as the plasma volume is reduced so the hematocrit value is increased.

Haematocrit value is decreased in:

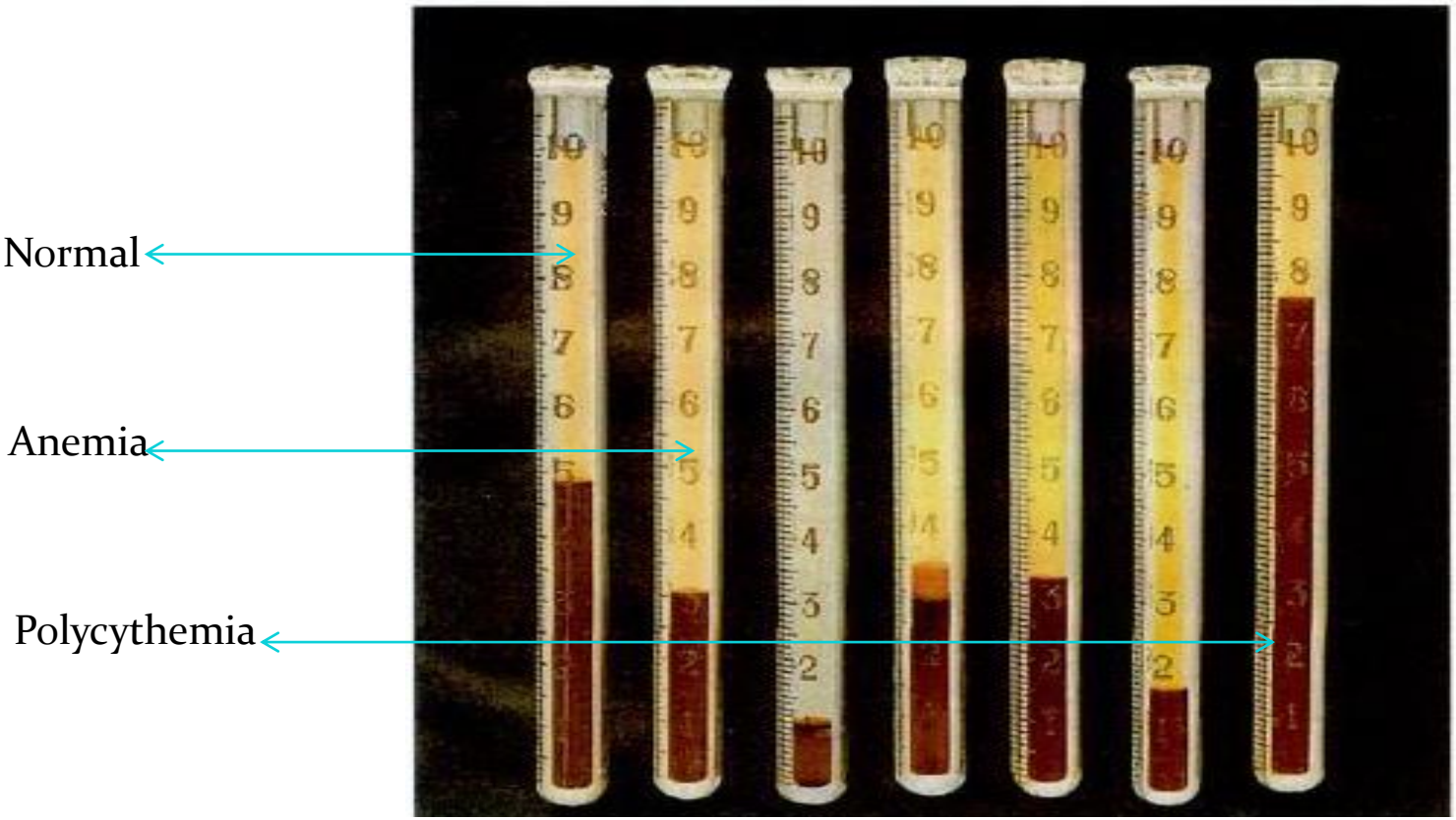
1. Anemia :

due to decreased number of RBCs, but its not used to determine type of anemia .

2. Overhydration:

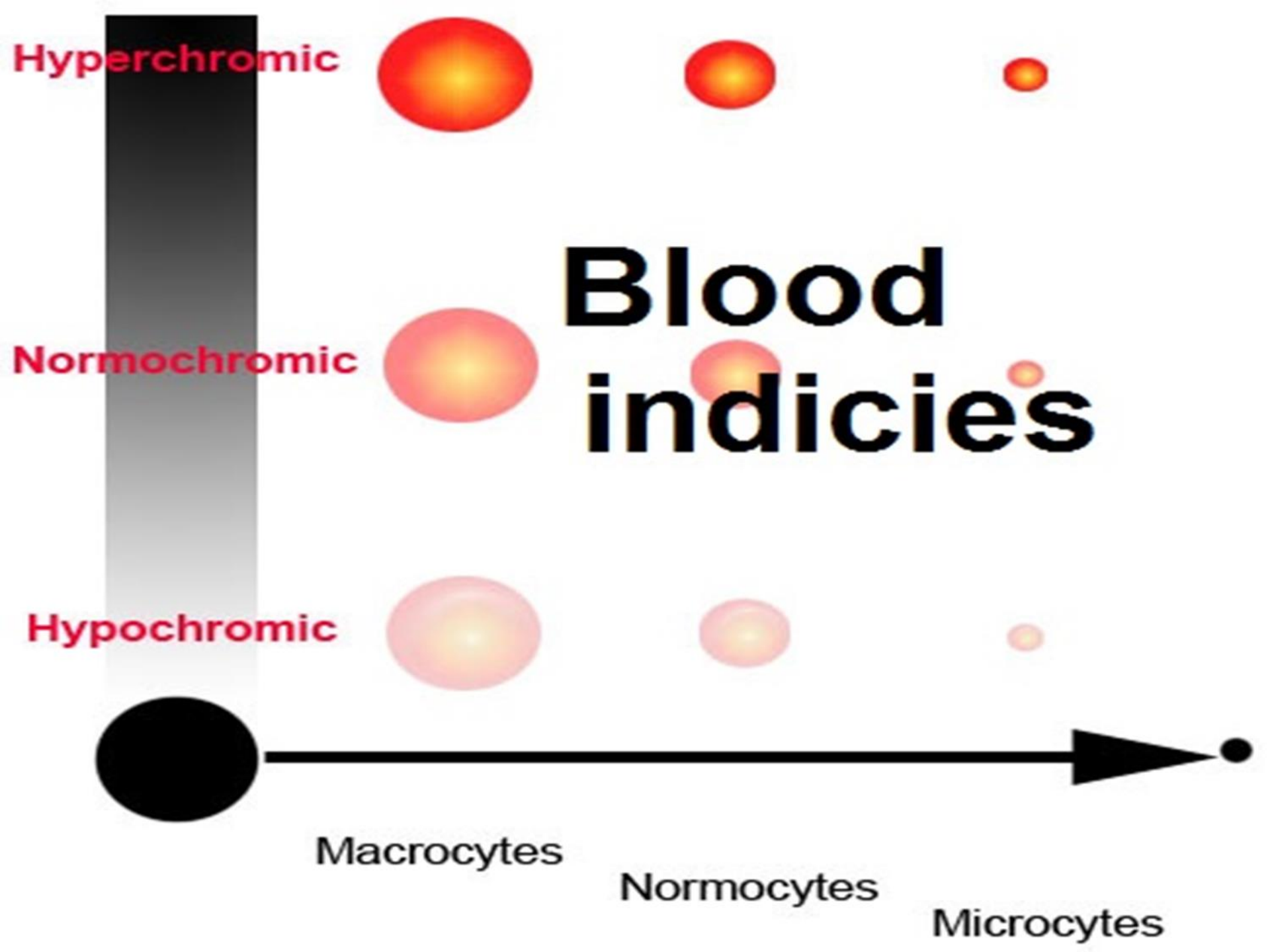
leading to increase plasma volume and so haematocrit value drops .

Haematocrit value may also be measured using **Wintrobe** tube, which is a bigger graduated tube the HV can be directly read after centrifugation.



Haematocrit value checklist

	Points of Evaluation	*Weight of step step (in marks)*	Marks of the student
1	The student held the hematocrit tube against the hematocrit scale, so that the bottom of the column of the red cells is aligned with the horizontal zero line (Exclude the height of clay)	1	
2	The student moved the tube across the scale until the line marked 100 passes through the top of the plasma column	1	
3	The student determined the value of packed cell volume correctly.	1	
Total marks		3	


Hyperchromic
Normochromic
Hypochromic
Blood indicies
Macrocytes
Normocytes
Microcytes

BLOOD INDICES

They are calculations that provide information about haemoglobin content and size of RBCs which aid in diagnosis and classification of anemia.

They include:

1. Mean corpuscular volume **(MCV)**
2. Mean corpuscular haemoglobin **(MCH)**
3. Mean corpuscular haemoglobin concentration **(MCHC)**

Mean corpuscular volume (MCV)

It is the average size of a RBC.

It is calculated by dividing HV by RBCs count.

$$\text{MCV} = \frac{\text{H.V} \times 10}{\text{RBCs count in millions /µL}}$$

Normally : **80 - 95 fL (µm³)**

Microcyte **< 80 fL** e.g. iron deficiency anemia.

Macrocyte **> 95 fL** e.g. vitamin B12 deficiency anemia.

NB. Femtoliter = 10^{-15} liter

Mean corpuscular hemoglobin (MCH)

It is the average amount of haemoglobin per RBC.

It is calculated by dividing the haemoglobin by RBCs count

$$\text{MCH} = \frac{\text{Hb content (g/dL)} \times 10}{\text{RBCs count in millions/µL}}$$

Normally : **27: 33 pg**

NB . Picogram = 10^{-12} g

Mean corpuscular haemoglobin concentration (MCHC)

It is the average Hb concentration in a single RBC.

It Is calculated by dividing the haemoglobin by the hematocrit.

$$\text{MCHC} = \frac{\text{Hb content (g/dL)} \times 100}{\text{hematocrit}}$$

Normally : **33 %** → Normochromic

> 35 % → Hyperchromic anemia e.g. hereditary spherocytosis.

< 33 % → Hypochromic anemia e.g. iron deficiency anemia.

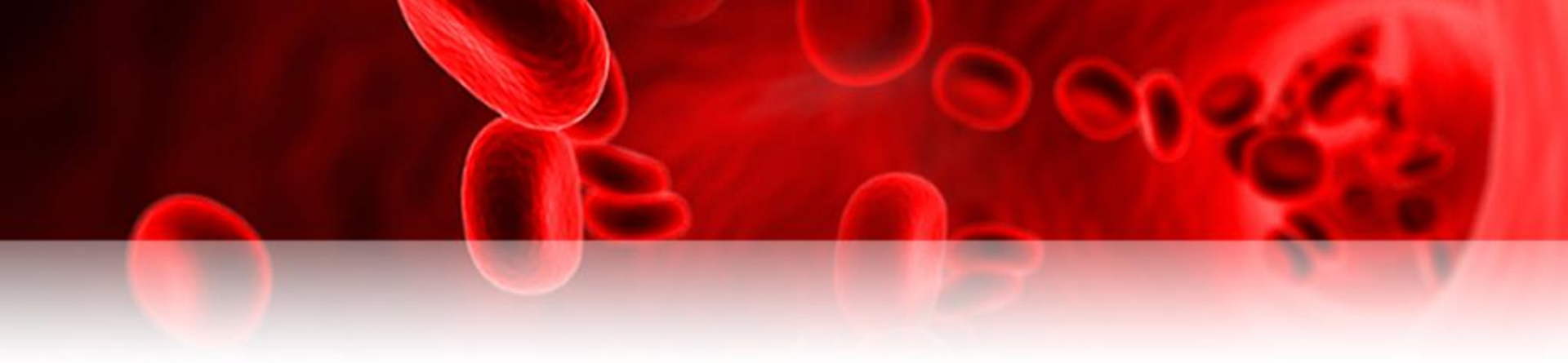
MCHC is not changed in vitamin B12 deficiency.

BLOOD INDICES

$$\text{MCV} = \frac{\text{H.V}}{\text{RBCs count in millions /µL}} \times 10$$

$$\text{MCH} = \frac{\text{Hb}}{\text{RBCs count in millions /µL}} \times 10$$

$$\text{MCHC} = \frac{\text{Hb}}{\text{H.V}} \times 100$$



Activity

Hemoglobin and Hematocrit

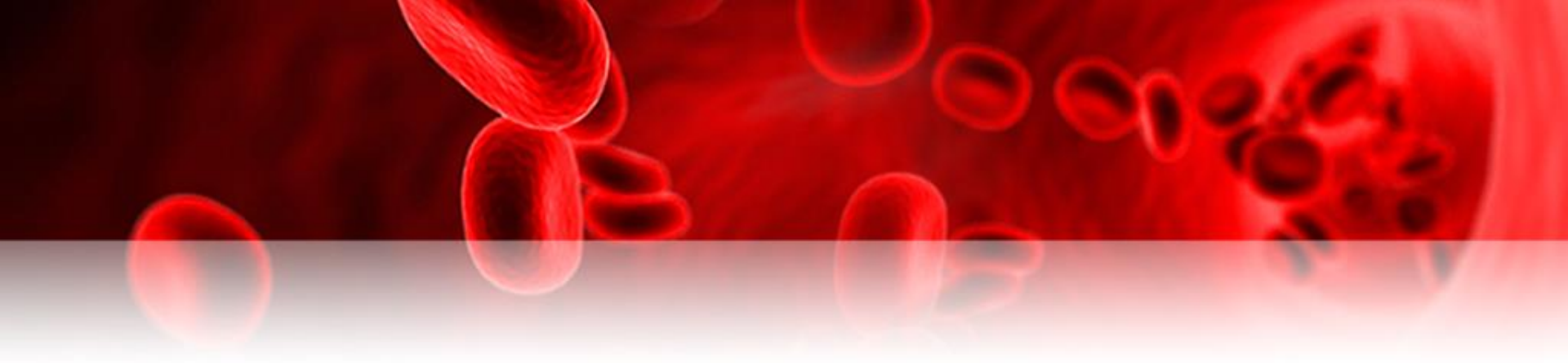
Q_1: What is the value of hemoglobin content of the blood sample in the dilution tube.

Is it normal, increased or decreased?

Q_2: State the hematocrit value of this capillary tube. State one cause of increased hematocrit value?

Blood groups

Q: Determine the blood group of this blood sample as regards both ABO and Rh systems. The person with this blood group has ……. antigen on his RBCs membrane.



Thank you